

Solvency Field Theory V12.4

Force-Sector and Electroweak Bridge from Worldline Framing

A bounded synthesis of the FORCE audit sequence

Shaun Fosmark

Independent researcher

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Mass is cadence; charge is worldline-framing holonomy; weak activity is closure-selected branch access; mass loading is the settlement bill for disturbing closure.

Publication boundary. This document claims architecture with boundaries, not premature completion. It distinguishes what SFT earns natively, what follows conditionally, what is mapped structurally, and what remains imported from the Standard Model. Reproducibility and provenance are identified by FORCE audit gates.

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1 The physical picture

A particle in Solvency Field Theory is not a point riding passively through a background spacetime. It is a real, maintained, internally active structure whose entire causal budget is spent at one speed — the speed of light. A massive particle “at rest” is not idle. It is cycling internally at its Compton frequency, $\omega_C = mc^2/\hbar$, pouring its full c -budget into the internal circulation that constitutes its mass. A photon, by contrast, spends that same budget entirely on spatial translation. Both move at c . The difference is where the motion goes.

This one-speed picture is not a metaphor. It is the relativistic four-velocity constraint $\beta_{\text{space}}^2 + \beta_{\text{phase}}^2 = 1$, rewritten as a statement about what the particle is *doing* rather than what it *has*. Mass is not a substance the particle carries. Mass is the rate at which the particle maintains itself — its internal heartbeat, its ticking.

The question this document addresses is: what else does the particle do besides tick?

1.1 Two faces of one worldline

Every massive particle has a worldline — a continuous history through four-dimensional spacetime. In SFT, that worldline is not an abstraction. It is a real, maintained structure: a tube in 4D, with spatial extent at the Compton scale, advancing forward at c , writing one Planck area of boundary history per tick. The tube has two independent geometric properties.

The first is its **pitch** — how tightly the helix is wound, how fast the internal circulation completes one cycle. This is the Compton frequency. This is mass. Two particles with different pitches have different masses. The pitch determines how fast the particle ticks, how much rest energy it carries, and how strongly it gravitates. This is the property we call ϕ_C , the Compton phase, and it is the source of everything in SFT’s gravity sector: the settlement lag, the metric closure, the Newtonian monopole, the MOND-like transition.

The second is its **twist** — how the transverse frame of the tube winds as the worldline advances. Picture the worldline as a garden hose. The water flows along the hose — that is the temporal advance, the ticking, the mass. But the hose itself can also be twisted around its own axis. One full twist, clockwise or counterclockwise. That twist is independent of the water flow. A fast-flowing hose and a slow-flowing hose can carry the same twist. A twisted hose and an untwisted hose can carry the same flow.

The twist is charge.

1.2 Why charge is not mass

This independence is not an assumption. It was forced by the FORCE audit sequence. FORCE 03 tested whether the Compton phase ϕ_C — the mass clock — could serve as the electromagnetic gauge phase. It cannot. Different particle species have different Compton frequencies (because they have different masses), but electromagnetism treats all unit charges identically regardless of mass. An electron and a proton have vastly different masses but exactly the same magnitude of electric charge. If charge were the mass clock, a proton’s electromagnetic behavior would differ from an electron’s by a factor of 1836. It does not.

FORCE 06 sharpened this into a theorem: any electromagnetic phase must lock universally across all charged species, independent of mass. The Compton phase fails this test because it runs at a mass-dependent rate. A separate phase — the twist, the worldline-framing angle, what we call θ_g — is mandatory.

The particle therefore carries two independent internal clocks on one worldline:

$$\boxed{\phi_C \rightarrow \text{mass} / \text{rest-clock cadence}} \quad (1)$$

$$\boxed{\theta_g \rightarrow \text{charge} / \text{worldline-framing holonomy}} \quad (2)$$

Mass is how fast the history ticks. Charge is how the history is twisted.

1.3 The gauge principle from geometry

The worldline tube has a transverse cross-section — a circle. That circle has an angular coordinate. But *where* you start measuring the angle is arbitrary. There is no preferred zero mark on the circumference of a tube. You can rotate your protractor without changing anything physical.

This is gauge invariance. Not imposed as a postulate. Arising from the geometry of the worldline itself: the local zero-angle of the transverse frame is unobservable.

The moment you demand that this freedom holds independently at every point along the worldline — that observers at different spacetime locations can independently choose their zero-angle — you are forced to introduce a connection field A_μ that compensates for the mismatch between neighboring choices. That connection transforms as $A_\mu \rightarrow A_\mu - g^{-1}\partial_\mu\alpha$ under a local rephasing $\theta_g \rightarrow \theta_g + \alpha(x)$. The gauge-invariant object built from A_μ is the field-strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. If the energy cost of the connection's curvature takes the Maxwell form $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, the Coulomb $1/r$ potential follows from the static limit.

This is the conditional electromagnetic bridge: *if* the worldline-framing connection carries Maxwell-form curvature cost, SFT recovers electrostatics from the geometry of the particle's own tube. The “if” is real — the curvature cost is admitted, not derived — and the document marks it honestly. But the route from worldline geometry to the Coulomb kernel is not imported from an abstract fiber bundle. It is the consequence of the tube having a cross-section with an angular coordinate whose origin is locally arbitrary.

1.4 Why history is rigid

A worldline is written history. Every tick inscribes one more Planck area on the boundary — the accumulated, unoverwritable record of everything that has happened. The past is not merely inaccessible. It is consumed. The spatial point you occupied one tick ago has dissolved; the temporal moment has been spent. Neither can be revisited, rewritten, or reframed.

This rigidity has a profound consequence for forces. Suppose a particle wants to change its branch state — to undergo a weak interaction, flipping from one identity to another. The particle's entire prior history says “I was framed *this* way.” The transition demands a new framing. But the old framing is already on the boundary. It cannot be erased. The contradiction between the new framing and the old costs energy — proportional to how much written history is being contradicted.

That cost is what SFT identifies as the physical origin of the Higgs mechanism.

In the Standard Model, the Higgs field has a nonzero vacuum expectation value, and disturbing it costs energy. The “Mexican hat” potential is drawn but never explained — why should the vacuum prefer a nonzero value? In SFT, the answer is that there is no such thing as an unframed worldline. Every particle that exists has been writing its frame to the boundary since it was created. The vacuum is not at zero because zero would mean “no history,” and every particle has a history. The Higgs potential is the cost function of historical rigidity, and mass is the ongoing bill for maintaining a frame orientation that was set by the past.

The photon pays nothing. It has no rest frame, no internal cycling, no historical branch commitment. Its worldline carries the twist — the electromagnetic connection propagating freely — but no Compton clock, no frame to maintain, no rigidity bill. Mass zero. Electromagnetic gauge carrier. Pure gauge excitation, unburdened by history.

The W and Z bosons pay a great deal. They mediate branch-changing transitions — their function is to rewrite framing. Every such rewriting pushes against the rigidity of already-written history. The mass they carry is the settlement cost of their role: bending what has already been inscribed.

1.5 Why the universe is handed

The Standard Model states that only left-handed particles couple to the weak force. It does not say why. Parity violation is one of the deepest unexplained asymmetries in physics, inserted into the Lagrangian as a structural fact with no dynamical origin.

SFT offers a candidate origin. The pre-shatter unified process — the single state that existed before the phase transition that created individual particles — had a closure orientation. A handedness. When it shattered into 10^{80} independent worldlines, those worldlines inherited the handedness of the state they came from. The primordial closure orientation became a fossil, embedded in every particle, preserved because the particles must collectively conserve the total properties of the state that produced them.

The weak force can only operate in the direction compatible with that fossil orientation. It is a ratchet set by the early universe. Left-handed particles are aligned with the ratchet; they can access the weak branch. Right-handed particles are against it; they cannot. Right-handed *antiparticles* can, because the antiparticle's boundary-address orientation flips the effective alignment. The selector is $s = \epsilon_U \eta_C \chi$, where ϵ_U is the primordial closure orientation, η_C is the particle/antiparticle sign, and χ is the chirality. Parity violation is not a rule imposed on the physics. It is a fossil of the shatter, inherited and conserved.

1.6 Charge as a network address

Every worldline writes to the boundary — the shared, unoverwritable record at $t = 0$. Each particle's write must connect to the boundary somewhere. The topology of that connection — how the worldline is plugged in — determines the particle's charge.

There are only so many topologically distinct ways to connect. You can wind your connection clockwise around the port (+1), counterclockwise (−1), or connect straight through with no winding (0). That is charge quantization: not from an abstract rule, but from the finite number of distinct connection topologies.

For quarks, the boundary has ports that accept three connections bundled together, where each cable carries 1/3 of the winding. The individual cables have fractional addresses, but only the bundled triple presents a valid complete connection to the rest of the network. An unbundled 1/3 connection is not a recognized address — the boundary protocol rejects it. That is confinement. Not a force holding quarks together, but a topological requirement: incomplete connections are not valid.

The integer write-address encoding replaces the Standard Model's fractional hypercharges with integer winding numbers $n_Y = 3Y$, so that all gauge quantum numbers are integers — topological winding counts on compact phases. The fractional charges of quarks are not fundamental fractions but projections of integer windings through the $Q = B_3 + n_Y/6$ charge formula. The 1/6 is a conversion factor, not a fractional quantum number.

1.7 The architecture that follows

With these physical pictures in hand, the rest of this document is their precise mathematical articulation.

Section 2 classifies every force sector as native, conditional, role-mapped, or imported. Sections 3–5 develop the three native sectors: inertia, gravity, and lag-stress. Section 6 establishes the two-phase identity and the worldline-framing origin of θ_g . Section 7 derives the conditional electromagnetic bridge from the framing connection through to the Coulomb kernel. Section 8 addresses the charge spectrum and the integer-winding encoding. Section 9 develops the weak-branch architecture, including the non-Abelian curvature and the closure-handedness selector. Section 10 treats budget-loading, the photon/ Z mass split, fermion masses, and the neutrino bridge. Section 11 states the Standard Model interface. Section 12 presents the full FORCE audit rollup and the final claim table.

Every section distinguishes what is earned from what is conditional, and what is conditional from what is open. The document claims architecture with boundaries. It does not claim completion.

2 Force-sector classification

The force definition used throughout this document is:

A force is the local observable response of real one-speed structures reallocating causal budget under compatibility and settlement constraints.

The current force map, after twenty audited gates:

Sector	SFT status	Claim level
Inertia / budget reallocation	Native candidate	Principle target
Gravity / settlement geometry	Native strongest	Active claim
Lag residual / settlement stress	Native phenomenological	Parameterized route
Gauge principle	Conditional theorem	If θ_g exists
Electromagnetic $U(1)$	Conditional route	Coulomb kernel conditional
Charge spectrum	Topology-motivated	Encoding, not derivation
Weak force	Branch-curvature bridge	Algebraic bridge
Strong force / QCD $\alpha \approx 1/137$	Imported SM Open	Role map only Response program

The condensed statement:

Native: gravity, inertia/budget, lag/settlement stress.

Conditional: native $U(1)$ if worldline-framing holonomy and Maxwell-form curvature cost are earned.

Role-mapped: electroweak mixing, Higgs/budget-loader, charge topology.

Imported: Standard Model gauge dynamics in precision practice.

3 Native sector I: inertia as budget reallocation

The one-speed budget is:

$$c^2 = v_{\text{external}}^2 + v_{\text{internal}}^2. \quad (3)$$

At rest, $v_{\text{internal}} = c$. When external motion is introduced, the budget cannot be spent twice. External velocity is paid for by reallocating internal maintenance capacity.

Inertia is resistance of maintained internal c -structure to external budget reallocation.

The full force-law recovery $F = dp/dt$ from one-speed closure remains an open theorem target. The native status is the architecture, not a finished derivation of all inertial dynamics.

4 Native sector II: gravity as settlement geometry

Gravity is the coarse-grained settlement geometry of maintained one-speed structures. The scalar monopole of the helix lag field produces a universal, unsigned $1/r$ potential — exactly gravity, not charge.

The weak-field settlement notation gives clock and ruler factors:

$$F = e^{-\phi}, \quad B = e^{+\phi}, \quad FB = 1. \quad (4)$$

The metric proxy is:

$$ds^2 = -F^2 c^2 dt^2 + B^2 d\ell^2, \quad (5)$$

and the radial acceleration is:

$$a_r = -c^2 \frac{d \ln F}{dr} = c^2 \frac{d\phi}{dr}. \quad (6)$$

The scalar channel sorted into gravity, not electromagnetism. This is why the FORCE audit's first result was a negative for EM and a confirmation of the gravity architecture.

Open items: derivation of G , root invariant, tensor dynamics, strong-field closure, compact-object limits, gravitational-wave consistency.

5 Native sector III: lag residual and settlement stress

The lag residual is the running difference between demanded and settled boundary capacity:

$$\Lambda(x, t) = A_{\text{demand}}(x, t) - A_{\text{settled}}(x, t). \quad (7)$$

The settlement dynamics are:

$$\partial_t \Lambda + \frac{\Lambda}{\tau_\Lambda} - D_\Lambda \nabla^2 \Lambda = \dot{A}_{\text{demand}}. \quad (8)$$

A scalar residual force density is $\mathbf{f}_{\text{scalar}} \sim -\nabla \Lambda$, and a tensor lag stress can be formed from the trace-free Hessian of Λ .

The combined force-density bookkeeping is:

$$f_i^{\text{total}} = f_i^{\text{SM}} + f_i^{\text{grav/settlement}} + f_i^{\text{lag}}. \quad (9)$$

This sector is native SFT language at the phenomenological level. Coefficient derivation from the action or root mechanics remains open.

6 The gauge problem and worldline-framing identity

The FORCE audits established that the electromagnetic gauge phase cannot be the Compton phase, cannot be chirality alone, and cannot be the relative phase between two particles. The viable SFT-native candidate is a conserved compact worldline-framing holonomy:

$$\theta_g = \text{compact worldline-framing} / \text{boundary-address holonomy}. \quad (10)$$

The gauge-invariant combination is:

$$H = d\theta_g + \omega_N, \quad (11)$$

where ω_N is the normal-frame connection of the worldline tube. Under local relabeling of the transverse frame:

$$\theta_g \rightarrow \theta_g + \alpha, \quad \omega_N \rightarrow \omega_N - d\alpha. \quad (12)$$

The combination $d\theta_g + \omega_N$ is invariant.

Gauge freedom is freedom to choose the local zero-angle of the worldline's transverse frame.

The winding charge is:

$$Q_W = \frac{N_e}{2\pi} \oint (d\theta_g + \omega_N), \quad (13)$$

which is topological and independent of ϕ_C .

The no-smuggling audit (FORCE 13B) confirmed that the compact circle can be read as the transverse angular direction around the particle's own maintained worldline tube, not an abstract fiber imported from outside.

7 Electromagnetic bridge: connection, curvature, Coulomb kernel

Once θ_g is identified as a compact local framing variable, local relabeling requires a connection:

$$\theta_g(x) \rightarrow \theta_g(x) + \alpha(x), \quad A_\mu \rightarrow A_\mu - \frac{1}{g_\theta} \partial_\mu \alpha. \quad (14)$$

The covariant derivative and field strength are:

$$D_\mu = \partial_\mu + ig_\theta A_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (15)$$

If the connection carries Maxwell-form curvature cost:

$$\mathcal{S}_{\text{EM}} = -\frac{1}{4g_\theta^2} \int F_{\mu\nu} F^{\mu\nu} + \int A_\mu J^\mu, \quad (16)$$

the static variational limit gives Poisson's equation $-\nabla^2 \phi = \rho$, whose point-source solution is $\phi(r) \propto 1/r$ with force scaling $F(r) \propto 1/r^2$.

Coulomb shape is conditionally earned. The coupling strength α is not.

The masslessness of A_μ is protected by the unbroken local $U(1)$ redundancy: a mass term $m_A^2 A_\mu A^\mu$ would break gauge invariance. This is why FORCE 04's requirement of a massless connection is natural in the framing picture — unbroken local relabeling symmetry forbids photon mass.

8 Charge spectrum and the integer-winding encoding

Unit winding handles integer charges ($Q/e \in \mathbb{Z}$) — sufficient for leptons and complete hadrons but not for quarks. Quarks require $Q/e \in \frac{1}{3}\mathbb{Z}$, which demands either a fundamental $e/3$ precharge or full electroweak hypercharge structure.

The integer write-address encoding replaces fractional hypercharges with integer winding numbers $n_Y = 3Y$:

Field	T_3 or B_3	n_Y	Q
ν_L	$+1/2$	-3	0
e_L	$-1/2$	-3	-1
u_L	$+1/2$	$+1$	$+2/3$
d_L	$-1/2$	$+1$	$-1/3$
e_R	0	-6	-1
u_R	0	$+4$	$+2/3$
d_R	0	-2	$-1/3$
H	$\pm 1/2$	$+3$	$1, 0$

The charge formula is $Q = B_3 + n_Y/6$. All hypercharges become integer write-address labels. Fractional charges are projections of integer windings, not fundamental fractions.

Fractional addresses are not free endpoints. They are valid only inside closure-complete bundled ports — the boundary protocol rejects incomplete connections. This is the role-map candidate for confinement.

The $B - L$ map provides a second structural route: $Q = T_{3L} + T_{3R} + (B - L)/2$, where the lepton/quark offset is no longer arbitrary but reflects the matter-sector write-address class.

The charge spectrum is encoded, not derived. The n_Y pattern and $B - L$ assignments remain open.

A candidate for the number of generations comes from the normal bundle of a timelike worldline in 3+1 dimensions: $D_\perp = 4 - 1 = 3$ independent normal directions. Three generations may correspond to three normal deformation modes of a worldline history. This is conditional, not a completed family-dynamics derivation.

9 Weak-branch architecture and non-Abelian curvature

The weak-sector bridge begins with branch access. A two-state internal orientation space naturally supports $SU(2)$ -like generators:

$$T_a = \frac{\sigma_a}{2}, \quad [T_a, T_b] = i\epsilon_{abc}T_c. \quad (17)$$

The charged branch-changing directions are:

$$T_+ = \frac{\sigma_1 + i\sigma_2}{2}, \quad T_- = \frac{\sigma_1 - i\sigma_2}{2}. \quad (18)$$

These are algebraic candidates for W^+ and W^- : operators that raise or lower the branch state.

If the branch frame is local, the branch connection is $W_\mu = W_\mu^a T_a$, with curvature:

$$F_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g_W \epsilon^{abc} W_\mu^b W_\nu^c. \quad (19)$$

The final term — the non-Abelian self-interaction — appears because branch generators do not commute. Branch switches applied in different order give different results, and the curvature of the connection reflects that non-commutativity. This is the structural reason the weak force is fundamentally different from electromagnetism.

Non-Abelian self-interaction appears from branch non-commutativity.

The closure-handedness selector determines which particles access the weak branch:

$$s = \epsilon_U \eta_C \chi, \quad g_W^{\text{eff}} = g_0 \frac{1+s}{2}, \quad (20)$$

where ϵ_U is the primordial closure orientation, η_C is the particle/antiparticle boundary-address sign, and χ is the chirality. Left-handed particles and right-handed antiparticles have $s = +1$ and access the weak branch. The rest do not.

Weak activity is branch access selected by primordial closure handedness.

This is a role map, not a complete weak-force derivation. Physical weak dynamics, couplings, and precision tests remain open.

10 Budget-loading, photon preservation, and the mass bridge

The budget-loading bridge asks whether branch-connection modes can become massive while preserving one exactly massless photon-like mode.

The neutral mass-loading matrix has the form:

$$M_0^2 = \frac{v^2}{4} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}, \quad (21)$$

with eigenvalues:

$$m_A^2 = 0, \quad m_Z^2 = \frac{v^2}{4}(g^2 + g'^2), \quad m_W^2 = \frac{g^2 v^2}{4}. \quad (22)$$

One mode remains exactly massless: the photon. One becomes massive: the Z . The charged branch modes $W^\pm$ acquire mass from the same loading.

Budget-loading can structurally make W/Z -like modes massive while preserving one massless photon-like mode.

The physical interpretation: the photon survives because one neutral combination of the branch and address connections remains a pure relabeling direction — disturbing it does not contradict any written history. The Z and $W^\pm$ modes correspond to directions that *do* change the framing, and the cost of contradicting written history is their mass.

For fermion masses, the Standard Model writes $m_f = y_f v / \sqrt{2}$. The SFT role map is: y_f is the species-specific loader-overlap coefficient — how strongly each particle is plugged into the budget-loading mechanism. Heavier particles have more historical commitment per tick, paying a larger maintenance bill.

Neutrinos are neutral branch-history modes that couple minimally to the loader. They barely plug into the mass billing machine, which is why their masses are tiny. The weak-access basis need not be the loader-overlap basis, providing a natural home for neutrino mixing.

SFT can describe the mass bill and why the billing machine should turn on, but not yet the exact billing rate. The Higgs VEV v , Yukawa couplings y_f , and mixing matrices remain open.

11 The Standard Model interface

The public interface remains deliberately conservative:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_{\text{SFT}}[T_{\mu\nu}, J_{\text{ops}}, \text{maintenance cost}]. \quad (23)$$

The Standard Model supplies gauge dynamics, representations, measured constants, and loop physics. SFT supplies one-speed closure, settlement geometry, lag stress, and maintenance accounting — and now proposes candidate geometric identities for charge, weak access, and mass loading.

This interface is not a retreat. It is the disciplined boundary that prevents overclaiming while preserving the force-sector architecture SFT has earned.

The Standard Model supplies the local gauge-field machinery in precision practice. SFT accounts the maintenance burden and proposes native origins for selected roles.

12 FORCE audit rollup and final claim table

Gates	Result	Boundary
01–02	Scalar helix kernel audited	Sorted into gravity; EM not in scalar channel
03–08	Gauge-framing identity	$\phi_C \neq \theta_g$; worldline-framing holonomy
09–13B	Charge topology, holonomy	Endpoint/ $\mathbb{Z}_3/B-L$ topology; not derived
14–15	Maxwell-action bridge, Coulomb	Conditional; α open
16–17C	Charge topology, weak chirality	Fractional ports; closure-handedness selector
18–18F	α program	Stiffness/response structure; α not derived
19–19B	Weak branch curvature	$SU(2)$ -like algebra; non-Abelian term earned
20–20F	Budget-loading, neutrinos	γ/Z mass split; loader/neutrino candidates

12.1 Final claim boundary

Safe claims:

Mass is cadence; charge is worldline-framing/boundary-address holonomy; weak activity is branch access selected by primordial closure handedness; mass loading is the settlement bill for disturbing closure.

If the framing connection carries Maxwell-form curvature cost, SFT recovers the Coulomb $1/r$ kernel. SFT has a non-Abelian branch-curvature bridge for weak-sector architecture.

Budget-loading can structurally make W/Z -like modes massive while preserving one massless photon-like mode.

Conditional claims:

Three generations may correspond to three normal directions of a worldline history in 3+1D.
Neutrino tiny mass may arise from minimal loader coupling or neutral self-closure.

Unsafe claims (not earned):

SFT has not derived α , the Higgs VEV, the Weinberg angle, gauge couplings, Yukawa couplings, CKM/PMNS matrices, or precision electroweak constants.

Final V12.4 statement:

SFT has sockets, switches, wiring logic, and a mass-billing map. It does not yet have the exact factory settings.

13 Rejected routes

The force audit explicitly rejected five tempting shortcuts.

EM from scalar helix geometry: rejected. Scalar helix kernels give unsigned monopoles. That is gravity, not charge.

Charge as ordinary Compton phase: rejected. The Compton phase is mass-dependent. Charge is not.

Charge as chirality alone: rejected. Chirality is at most $\mathbb{Z}_2$. Electric charge requires compact $U(1)$ structure.

α from naive helix self-interaction ratios: rejected. The tested ratios were cutoff-dominated and did not produce $1/137$.

Linear confinement from local three-branch completeness: rejected. The tested model saturated instead of producing $E \sim \sigma L$.

Rejecting these routes is not failure. It is what prevents the theory from becoming vague.

Acknowledgments

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References

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